

On Prediction of Extreme Values

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This paper discusses the prediction of the extreme values of a stationary random process having an arbitrary bandwidth spectrum. Formulas to predict the extreme values are derived for a given number of observations and time by applying order statistics, and the effect of bandwidth of spectrum on the extreme values is discussed. The most probable extreme value and the extreme value for a preassigned probability of being exceeded are obtained. The estimation formulas are applied to predict the extreme value of the pitching motion and acceleration of a ship in irregular seas and waves generated in a towing tank. It is found that the extreme values observed in model experiments fall fairly well in the vicinity of the predicted most probable values. Theoretically, however, the chance that the extreme value will exceed the most probable value is rather high. Hence, the extreme value for a preassigned probability of being exceeded appears to be appropriate for ship (or marine structure) design consideration.

Introduction

FOR VARIOUS RANDOM PHENOMENA encountered in the fields of naval and ocean engineering, it is often necessary to predict the largest value (extreme value) of the phenomenon that is expected to occur in a certain period of time. For instance, prediction of the largest motions and wave-induced forces, etc., is essential for design of ships and marine structures.

Random phenomena dealt with in naval and ocean engineering are usually stationary Gaussian (normal) processes and have been treated as having narrow-band spectra. The spectra of many random phenomena observed in practice, however, cover a certain range of frequencies, and often they have several maxima and minima during one cycle as determined from zero crossings, as contrasted to a single peak and trough for a narrow-band random process.

A statistical study of the extreme values of random processes was first discussed by Rice [1],² and later the subject was studied in great detail by Cartwright and Longuet-Higgins [2], and Huston and Spokinski [3], independently, including the effect of bandwidth of the spectrum on the probability distribution. In particular, reference [2] discusses the extreme value of the maxima that is most likely to occur (the most probable extreme value) throughout the range of $-\infty$ to $+\infty$ in a certain number of observations. The chance that a value greater than the predicted extreme value will occur, however, is rather high; theoretically it is 0.632 for a random process having narrow-band spectrum. It appears, therefore, that the most probable extreme value is too low to be considered for engineering design purposes. In addition, it may be more meaningful to express the extreme value in terms of time rather than as a function of number of observations.

This paper discusses the prediction of the extreme value of the maxima of a random process that will occur in a certain period of time during which the sea environment is constant. The extreme value is estimated for an arbitrarily chosen small probability of being exceeded. For example, by choosing the probability of being exceeded as 0.01, one may predict the extreme value for which only one ship (or marine structure) in 100 sister ships (or structures) operating under statistically identical environment will experience a value higher than the predicted value in a specified period of time.

Formulas to estimate the extreme values are developed by applying order statistics, and the effect of bandwidth of the spec-

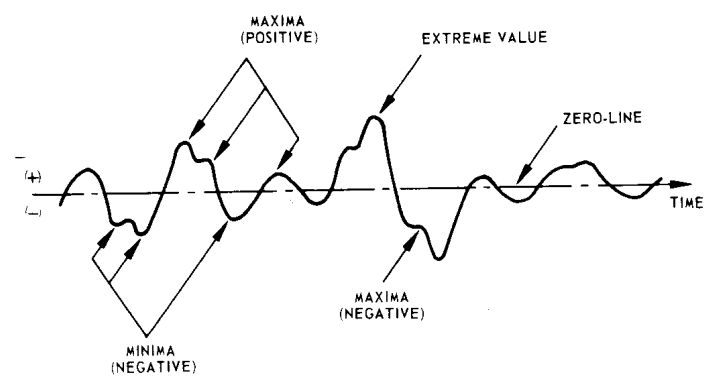


Fig. 1 Explanatory sketch of a random process $X(t)$

trum on the extreme values is discussed. It is shown that the extreme value expressed in terms of time is a function only of the area and the second moment of the spectral density function of a random process. The estimation formulas are applied to predict the extreme value of the pitching motion and acceleration of a ship in irregular seas and waves generated in a towing tank.

Probability distribution function of maxima with positive values

In order to obtain the extreme value of the maxima of a random process by applying order statistics, the probability distribution function of the maxima is derived. In particular, the maxima of the random process with positive values is considered in this study, since the maxima with negative values are unnecessary for engineering design purposes. For a better understanding of this issue, Fig. 1 was prepared.

Figure 1 is an explanatory sketch of a random process $X(t)$ for which the maxima could be anywhere in the range of $(-\infty, +\infty)$ and several maxima could occur during one cycle as defined by zero crossings. Here, the maxima of $X(t)$ are defined as peaks which satisfy the conditions $\dot{X}(t) = 0$ and $\ddot{X}(t) < 0$. The spectral density function of a process illustrated in the figure is not narrow-banded, in general; that is, a significant portion of the spectrum covers a range of frequencies. For this type of random process, the magnitudes of the maxima with positive values $\{X(t) > 0, \dot{X}(t) = 0, \ddot{X}(t) < 0\}$ or the minima with negative values $\{X(t) < 0, \dot{X}(t) = 0, \ddot{X}(t) > 0\}$ would be critical if they exceeded a certain value, and hence the statistical extreme values of these maxima and the minima provide valuable information for engineering design. Since the statistical properties of the minima with negative values are the same as those of the maxima

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² Numbers in brackets designate References at end of paper.

Manuscript received at SNAME Headquarters April 19, 1972. Revised manuscript received June 19, 1972.

with positive values, only the latter are considered in the following analysis.

The probability density function of the maxima with positive values of a random process $X(t)$ can be derived by applying the concept of the threshold crossing problem. That is, consider a random process $X(t)$ that is stationary and Gaussian with zero mean. Then, the joint probability density function of $X(t)$, $\dot{X}(t)$, and $\ddot{X}(t)$ can be written by

$$f(\mathbf{X}) = \frac{1}{(2\pi)^{3/2} |\boldsymbol{\Sigma}|^{1/2}} e^{-\frac{1}{2} \mathbf{X}' \boldsymbol{\Sigma}^{-1} \mathbf{X}} \quad (1)$$

where

$$\mathbf{X}' = (X(t) \quad \dot{X}(t) \quad \ddot{X}(t))$$

and the covariance matrix $\boldsymbol{\Sigma}$ can be obtained from the power spectral density function $S(\omega)$ of the process $X(t)$ as

$$\boldsymbol{\Sigma} = \begin{pmatrix} m_0 & 0 & -m_2 \\ 0 & m_2 & 0 \\ -m_2 & 0 & m_4 \end{pmatrix} \quad (2)$$

where

$$m_0 = \int_0^\infty S(\omega) d\omega$$

$$m_2 = \int_0^\infty \omega^2 S(\omega) d\omega$$

$$m_4 = \int_0^\infty \omega^4 S(\omega) d\omega$$

$S(\omega)$ = ordinate of power density spectrum of $X(t)$

ω = frequency, cps

The maxima of the process with positive values satisfy the condition

$$X(t) > 0, \dot{X}(t) = 0, \ddot{X}(t) < 0 \quad (3)$$

and the expected number of maxima above a specified level $X(t) = \xi$, denoted as $\bar{N}_\xi$, is obtained by

$$\bar{N}_\xi = E[N(\xi, t)] = \int_\xi^\infty \int_{-\infty}^0 |\ddot{x}(t)| f(x, 0, \ddot{x}) d\ddot{x} dx \quad (4)$$

The expected number of maxima with positive values, denoted as $\bar{N}_{m>0}$, then becomes

$$\bar{N}_{m>0} = E[N(x > 0, t)] = \int_0^\infty \int_{-\infty}^0 |\ddot{x}(t)| f(x, 0, \ddot{x}) d\ddot{x} dx \quad (5)$$

The ratio of these two expected values is approximately equivalent to the probability that the maximum value will exceed a level ξ ; that is, by letting the random variable Ξ be the maximum positive value,

$$Pr\{\Xi \geq \xi\} = E\left[\frac{N(\xi, t)}{N(x > 0, t)}\right] \sim \frac{E[N(\xi, t)]}{E[N(x > 0, t)]} = \frac{\bar{N}_\xi}{\bar{N}_{m>0}} \quad (6)$$

Then, the probability density function of the maxima is given by

$$f(\xi) = \frac{d}{d\xi} \left\{ 1 - \frac{\bar{N}_\xi}{\bar{N}_{m>0}} \right\} = \frac{\int_{-\infty}^0 |\ddot{x}(t)| f(\xi, 0, \ddot{x}) d\ddot{x}}{\int_0^\infty \int_{-\infty}^0 |\ddot{x}(t)| f(x, 0, \ddot{x}) d\ddot{x} dx} \quad (7)$$

The numerator in the equation (7) was obtained by Cartwright and Longuet-Higgins as [2]:

$$\begin{aligned} & \int_{-\infty}^0 |\ddot{x}(t)| f(\xi, 0, \ddot{x}) d\ddot{x} \\ &= \frac{1}{(2\pi)\sqrt{m_0}} \sqrt{\frac{m_4}{m_2}} \left[\frac{\epsilon}{\sqrt{2\pi}} \exp \left\{ -\frac{1}{2\epsilon^2} \left(\frac{\xi}{\sqrt{m_0}} \right)^2 \right\} \right. \\ & \quad \left. + \sqrt{1 - \epsilon^2} \left(\frac{\xi}{\sqrt{m_0}} \right) \exp \left\{ -\frac{1}{2} \left(\frac{\xi}{\sqrt{m_0}} \right)^2 \right\} \right. \\ & \quad \left. \times \left\{ 1 - \Phi \left(-\frac{\sqrt{1 - \epsilon^2}}{\epsilon} \frac{\xi}{\sqrt{m_0}} \right) \right\} \right] \quad (8) \end{aligned}$$

where

$$\epsilon = \sqrt{1 - \frac{m_2^2}{m_0 m_4}}$$

$$\Phi(u) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^u e^{-\frac{u^2}{2}} du$$

Note that ϵ in the equation (8) is the bandwidth parameter of the spectrum defined by Cartwright and Longuet-Higgins. For

Nomenclature

$f(x)$ = probability density function of random variable X	T = time, hr	$\tilde{\xi}_n$ = nondimensional most probable extreme value in n observations
$F(x)$ = cumulative distribution function of random variable X	$X(t)$ = random process	$\hat{\xi}_n$ = nondimensional extreme value in n observations for which the probability that the extreme value will exceed $\hat{\xi}_n$ is α
$g(\xi_n)$ = probability density function of largest value in n observations	$\mathbf{x}'$ = random vector $[x(t) \quad \dot{x}(t) \quad \ddot{x}(t)]$	H, η = random variable defined by $\Xi/\sqrt{m_0}$
m_0 = area under spectral density function of random process $X(t)$	α = probability that extreme value exceeds a specified value	$\oplus, \theta$ = random variable defined by $n\{1 - F(\xi_n)\}$
m_2 = second moment of spectral density function	γ_n = dimensional most probable extreme value	Ξ, ξ = random variable, maximum positive value of $X(t)$
m_4 = fourth moment of spectral density function	$\hat{\gamma}_n$ = dimensional extreme value in n observations for which the probability that the extreme value will exceed $\hat{\gamma}_n$ is α	$\boldsymbol{\Sigma}$ = covariance matrix of random vector $\mathbf{x}$
$\Delta = m_0 m_4 - m_2^2$	$\epsilon = \sqrt{1 - \frac{m_2^2}{m_0 m_4}}$	$\Phi(u)$ = cumulative distribution function of the standardized normal distribution
n = number of observations	Z, ζ = ordered random variable	ω = frequency
$\bar{N}_\xi$ = expected number of maxima above a specified level	Z_n, ζ_n = largest ζ in n ordered sample; i.e., extreme value in n observations	
$\bar{N}_{m>0}$ = expected number of maxima with positive value		
$S(\omega)$ = power density spectrum		

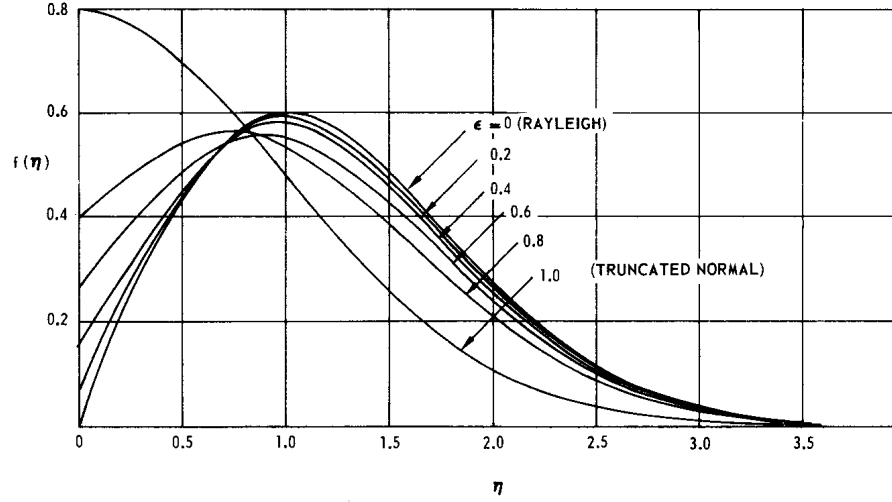


Fig. 2 Probability density function of random variable H as a function of bandwidth parameter ϵ

a random process having a narrow-band spectrum, ϵ is equal to zero, and Ξ has a Rayleigh probability distribution. For a wide-band spectrum ϵ is equal to 1 and Ξ has a normal distribution [2].

The denominator of Equation (7) is evaluated in the Appendix. It is given by

$$\begin{aligned} \bar{N}_{m>0} &= \int_0^\infty \int_{-\infty}^0 |\ddot{x}(t)| f(x, 0, \ddot{x}) d\ddot{x} dx \\ &= \frac{1}{4\pi} \left(\sqrt{\frac{m_2}{m_0}} + \sqrt{\frac{m_4}{m_2}} \right) = \frac{1}{4\pi} \left(\frac{1 + \sqrt{1 - \epsilon^2}}{\sqrt{1 - \epsilon^2}} \right) \sqrt{\frac{m_2}{m_0}} \quad (9) \end{aligned}$$

Thus, from equations (7), (8), and (9), the probability density function becomes

$$\begin{aligned} f(\xi) &= \frac{\left(\frac{2}{\sqrt{m_0}} \right)}{1 + \sqrt{1 - \epsilon^2}} \left[\frac{\epsilon}{\sqrt{2\pi}} \exp \left\{ -\frac{1}{2\epsilon^2} \left(\frac{\xi}{\sqrt{m_0}} \right)^2 \right\} \right. \\ &\quad \left. + \sqrt{1 - \epsilon^2} \left(\frac{\xi}{\sqrt{m_0}} \right) \exp \left\{ -\frac{1}{2} \left(\frac{\xi}{\sqrt{m_0}} \right)^2 \right\} \right] \\ &\quad \times \left\{ 1 - \Phi \left(-\frac{\sqrt{1 - \epsilon^2}}{\epsilon} \frac{\xi}{\sqrt{m_0}} \right) \right\} \quad 0 \leq \xi < \infty \quad (10) \end{aligned}$$

If the random variable Ξ is expressed in a nondimensional form $\Xi/\sqrt{m_0} = H$, then the probability density function of H becomes

$$\begin{aligned} f(\eta) &= \frac{2}{1 + \sqrt{1 - \epsilon^2}} \left[\frac{\epsilon}{\sqrt{2\pi}} \exp \left\{ -\frac{\eta^2}{2\epsilon^2} \right\} \right. \\ &\quad \left. + \sqrt{1 - \epsilon^2} \eta e^{-\frac{\eta^2}{2}} \left\{ 1 - \Phi \left(-\frac{\sqrt{1 - \epsilon^2}}{\epsilon} \eta \right) \right\} \right] \\ &\quad 0 \leq \eta < \infty \quad (11) \end{aligned}$$

In case the random process $X(t)$ has a narrow-band spectrum ($\epsilon = 0$)

$$f(\eta) = \eta e^{-\frac{\eta^2}{2}} \quad 0 \leq \eta < \infty \quad (12)$$

which is the Rayleigh distribution, while if $X(t)$ has a wide-band spectrum ($\epsilon = 1$)

$$f(\eta) = \sqrt{\frac{2}{\pi}} e^{-\frac{\eta^2}{2}} \quad 0 \leq \eta < \infty \quad (13)$$

which is a truncated normal distribution (truncated at $\eta = 0$). Figure 2 shows the probability density function of H for various ϵ -values.

The cumulative distribution functions of Ξ and H are derived simply by integrating equations (10) and (11) with respect to ξ and η , respectively. They are given by

$$\begin{aligned} F(\xi) &= \frac{2}{1 + \sqrt{1 - \epsilon^2}} \left[-\frac{1}{2} (1 - \sqrt{1 - \epsilon^2}) + \Phi \left(\frac{\xi}{\epsilon \sqrt{m_0}} \right) \right. \\ &\quad \left. - \sqrt{1 - \epsilon^2} \exp \left\{ -\frac{1}{2} \left(\frac{\xi}{\sqrt{m_0}} \right)^2 \right\} \right. \\ &\quad \left. \times \left\{ 1 - \Phi \left(-\frac{\sqrt{1 - \epsilon^2}}{\epsilon} \frac{\xi}{\sqrt{m_0}} \right) \right\} \right] \quad 0 \leq \xi < \infty \quad (14) \end{aligned}$$

and

$$\begin{aligned} F(\eta) &= \frac{2}{1 + \sqrt{1 - \epsilon^2}} \left[-\frac{1}{2} (1 - \sqrt{1 - \epsilon^2}) + \Phi \left(\frac{\eta}{\epsilon} \right) \right. \\ &\quad \left. - \sqrt{1 - \epsilon^2} \exp \left\{ -\frac{\eta^2}{2} \right\} \left\{ 1 - \Phi \left(-\frac{\sqrt{1 - \epsilon^2}}{\epsilon} \eta \right) \right\} \right] \\ &\quad 0 \leq \eta < \infty \quad (15) \end{aligned}$$

Extreme values

In this section, the extreme value of the maxima will be derived by applying order statistics. The extreme value defined here is the largest value of the maxima that will occur in n observations (see Fig. 1).

Let $(\eta_1, \eta_2, \dots, \eta_n)$ be a random sample of size n , where η_i is the observed maxima of a random process $X(t)$ in nondimensional form. All η_i have the same probability density function given in equation (11). Let $(\zeta_1, \zeta_2, \dots, \zeta_n)$ be an ordered sample of η_i with $\zeta_1 < \zeta_2 < \dots < \zeta_n$. Then, each ζ_i can be considered as the outcome of the random variable Z_i , which has its own probability density function different from $f(\eta)$. For example, the random variable Z_n , the largest ζ in the ordered sample, has the following probability density function:

$$g(\zeta_n) = n f(\zeta_n) \{F(\zeta_n)\}^{n-1} \quad (16)$$

where

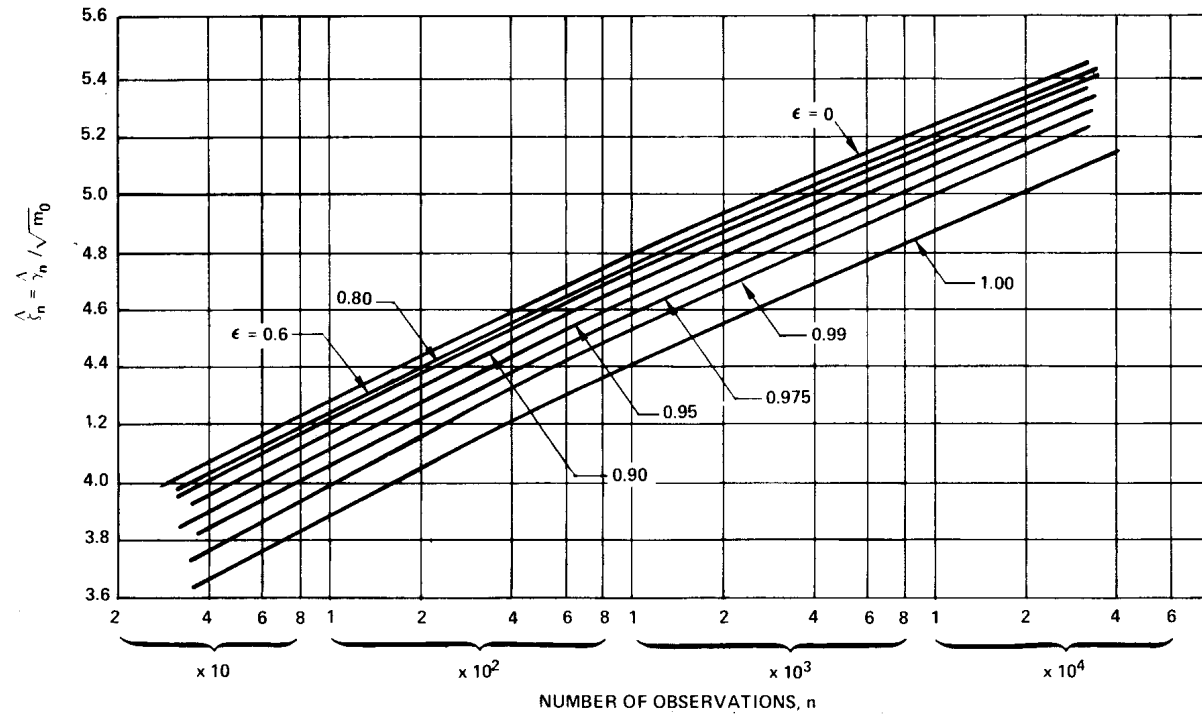


Fig. 3 Extreme values $\hat{\zeta}_n$ (amplitude) as function of number of observations for various bandwidth parameters, ϵ ($\alpha = 0.01$)

$g(\zeta_n)$ = probability density function of largest value in n observations

$f(\zeta_n)$ = probability density function given in equation (11) where η is ζ_n

$F(\zeta_n)$ = cumulative distribution function given in equation (15) where η is ζ_n

n = number of observations

The value of ζ_n for which the probability density function $g(\zeta_n)$ becomes maximum is the extreme value that is most likely to occur in n observations, and it is called the "most probable extreme value," denoted by $\hat{\zeta}_n$. Its value can be obtained by letting the derivative of $g(\zeta_n)$ be zero. It should be noted, however, that the probability that the extreme value will exceed $\hat{\zeta}_n$ is quite large, as will be shown later. It is highly desirable, therefore, to predict the extreme value for which the probability of being exceeded is very small. In other words, let us choose a very small number, α , and obtain the extreme value $\hat{\zeta}_n$ for which the following relationship holds:

$$Pr\left\{\begin{array}{l} \text{Extreme value of maxima} \\ \text{in } n \text{ observations} \end{array} > \hat{\zeta}_n\right\} = \alpha \quad (17)$$

Thus, from equations (16) and (17):

$$\int_0^{\hat{\zeta}_n} g(\zeta_n) d\zeta_n = \{F(\hat{\zeta}_n)\}^n = 1 - \alpha \quad (18)$$

Considering that α is small and n is large, we have

$$F(\hat{\zeta}_n) = (1 - \alpha)^{\frac{1}{n}} \sim 1 - \frac{\alpha}{n} + o(\alpha^2) \quad (19)$$

Then, from equations (15) and (19):

$$\begin{aligned} \sqrt{1 - \epsilon^2} \exp\left\{-\frac{\hat{\zeta}_n^2}{2}\right\} \left\{1 - \Phi\left(-\frac{\sqrt{1 - \epsilon^2}}{\epsilon} \hat{\zeta}_n\right)\right\} \\ + \left\{1 - \Phi\left(\frac{\hat{\zeta}_n}{\epsilon}\right)\right\} = \left(\frac{1 + \sqrt{1 - \epsilon^2}}{2}\right) \left(\frac{\alpha}{n}\right) \end{aligned} \quad (20)$$

The $\hat{\zeta}_n$ that can be evaluated from the equation (20) for a given α is the desired extreme value (nondimensional) of the maxima of a random process which has a bandwidth parameter ϵ . The dimensional extreme value, denoted by $\hat{\gamma}_n$, is given by $\hat{\zeta}_n \sqrt{m_0}$ for the amplitude of the random process.

It should be noted that for a random process having a bandwidth parameter, ϵ , less than 0.90, equation (20) can be drastically simplified as follows; that is, for $\epsilon \leq 0.9$,

$$\Phi\left(-\frac{\sqrt{1 - \epsilon^2}}{\epsilon} \hat{\zeta}_n\right) \sim 0 \quad (21)$$

$$\Phi\left(\frac{\hat{\zeta}_n}{\epsilon}\right) \sim 1$$

Hence, equation (20) becomes

$$\sqrt{1 - \epsilon^2} \exp\left\{-\frac{\hat{\zeta}_n^2}{2}\right\} = \left(\frac{1 + \sqrt{1 - \epsilon^2}}{2}\right) \left(\frac{\alpha}{n}\right) \quad (22)$$

The extreme value, $\hat{\zeta}_n$, for small α is then derived as

$$\hat{\zeta}_n = \sqrt{2 \ln \left(\frac{\sqrt{1 - \epsilon^2}}{1 + \sqrt{1 - \epsilon^2}} \cdot \frac{2n}{\alpha} \right)} \quad \text{for } \epsilon \leq 0.9 \quad (23)$$

For a random process having a narrow-band spectrum ($\epsilon = 0$), the nondimensional extreme value, $\hat{\zeta}_n$, is obtained from equation (23) as

$$\hat{\zeta}_n = \sqrt{2 \ln \frac{n}{\alpha}} \quad \text{for small } \alpha \quad (24)$$

The dimensional extreme value of the amplitude of a process, $\hat{\gamma}_n$, is then given by

$$\hat{\gamma}_n = \sqrt{2 \ln \frac{n}{\alpha}} \sqrt{m_0} \quad (25)$$

Figure 3 shows the extreme value $\hat{\zeta}_n$ for $\alpha = 0.01$ calculated

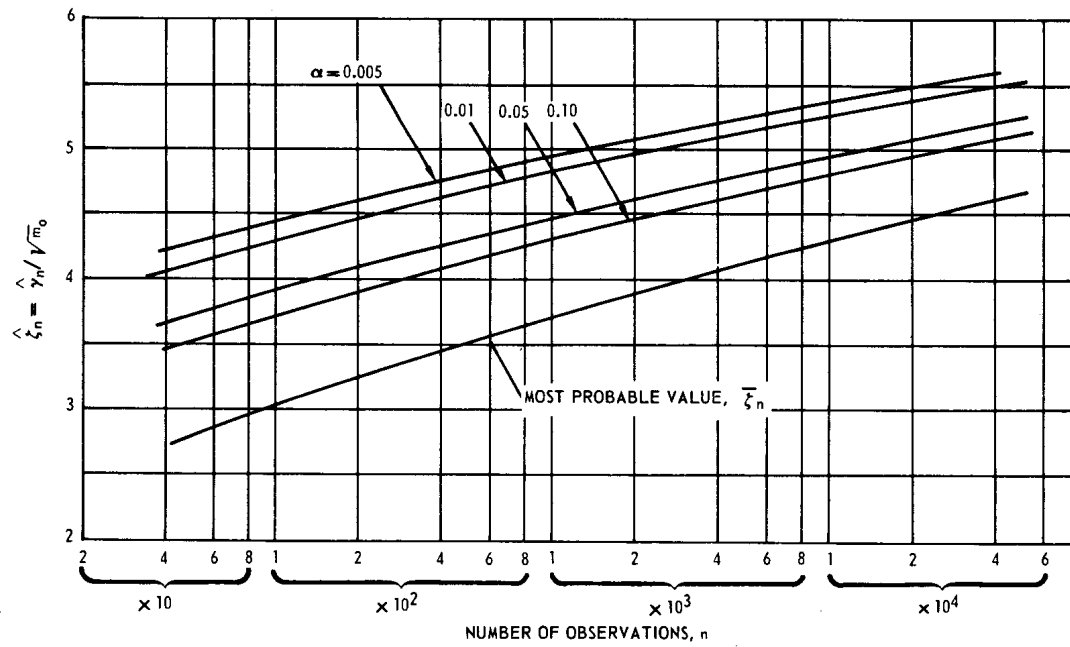


Fig. 4 Extreme values $\hat{\zeta}_n$ (amplitude) for the Rayleigh distribution ($\epsilon = 0$) as a function of probability of being exceeded α

from equations (20) and (23) as a function of the band width parameter ϵ . As can be seen in the figure, there is no significant difference in the $\hat{\zeta}_n$ -values for ϵ up to 0.9. The difference in $\hat{\zeta}_n$ -values is noticeable for ϵ greater than 0.90. It is noted that Fig. 3 gives the impression that $\hat{\zeta}_n$ decreases with increase of the bandwidth parameter. This is true for the same number of cycles, n . For a given period of time, however, this trend does not hold. It will be shown in the following section that the extreme value $\hat{\zeta}_n$ expressed in terms of time is no longer a function of ϵ , but is a function of the area and the second moment of the spectral density function of the process.

It may be of interest to discuss in more detail the extreme values pertaining to the Rayleigh distribution ($\epsilon = 0$). First, a comparison between the extreme value, $\hat{\zeta}_n$, and the most probable extreme value, $\bar{\zeta}_n$, is discussed. The most probable extreme value applicable to the Rayleigh distribution was derived by Longuet-Higgins in 1952, who first introduced the application of the Rayleigh distribution to naval engineering [4]. His result is for large n :

$$\begin{cases} \bar{\zeta}_n = \sqrt{2 \ln n} \\ \bar{\gamma}_n = \sqrt{2 \ln n} \sqrt{m_0} \quad \text{for amplitude} \end{cases} \quad (26)$$

The most probable extreme value $\bar{\zeta}_n$ may be interpreted as being most likely to occur since it is the value for which the probability density function of the random variable Z_n has a peak. The chance of occurrence of a value greater than $\bar{\zeta}_n$, however, is rather high. That is, from equations (16) and (26), the probability that the extreme value will exceed $\bar{\zeta}_n$ becomes for large n

$$\begin{aligned} & \lim_{n \rightarrow \infty} \Pr\{\text{Extreme value} > \bar{\zeta}_n\} \\ &= \lim_{n \rightarrow \infty} \int_{\bar{\zeta}_n}^{\infty} g(\zeta_n) d\zeta_n = \lim_{n \rightarrow \infty} [1 - F(\bar{\zeta}_n)] \\ &= \lim_{n \rightarrow \infty} \left\{ 1 - \left(1 - \frac{1}{n} \right)^n \right\} = 1 - e^{-1} = 0.632 \end{aligned} \quad (27)$$

It is noted that $F(\bar{\zeta}_n)$ in equation (27) is obtained by integrating equation (12) from zero to $\bar{\zeta}_n$.

Since this probability is very large, it may not be most appropriate to choose $\bar{\zeta}_n$ for prediction. On the other hand, an extreme value $\hat{\zeta}_n$ is determined for a desired probability of being exceeded. Although the decision concerning which probability value should be taken in practice depends on the designer's discretion, the probability of 0.01 appears to be appropriate. This subject is discussed in detail later in connection with the example of application.

The extreme values $\hat{\zeta}_n$ for the Rayleigh distribution for $\alpha = 0.005, 0.01, 0.05$, and 0.10 together with the most probable value $\bar{\zeta}_n$ are shown in Fig. 4.

The most probable extreme value of a random process having an arbitrary bandwidth parameter, ϵ , can be derived from equation (16); that is, by letting the derivative of equation (16) with respect to ζ_n be zero, one has

$$f'(\zeta_n)F(\zeta_n) + (n-1)\{f(\zeta_n)\}^2 = 0 \quad (28)$$

Since $f(\zeta_n)$ and $F(\zeta_n)$ are given as a function of ϵ in equations (11) and (15), respectively, the solution of the equation (28) yields the most probable extreme value $\bar{\zeta}_n$ for a given ϵ . Although an analytical formula to evaluate $\bar{\zeta}_n$ applicable for all ϵ cannot be developed due to the complexity of equation (28), a simple formula may be derived for large n and for $\epsilon \leq 0.9$ by using the approximation given in equation (21). That is, by neglecting terms of small order of magnitude, equation (28) becomes

$$-\frac{\sqrt{1-\epsilon^2}(1+\sqrt{1-\epsilon^2})}{2} e^{-\frac{\zeta_n^2}{2}} + n(1-\epsilon^2)e^{\zeta_n^2} + o(\zeta_n) = 0 \quad (29)$$

Then, the following solution can be obtained from the equation (29):

$$\bar{\zeta}_n = \sqrt{2 \ln \left\{ \frac{2\sqrt{1-\epsilon^2}}{1+\sqrt{1-\epsilon^2}} n \right\}} \quad \text{for } \epsilon \leq 0.9 \quad (30)$$

Note that for a narrow-band random process ($\epsilon = 0$), equation (30) agrees with the result given by Longuet-Higgins [see equation (26)].

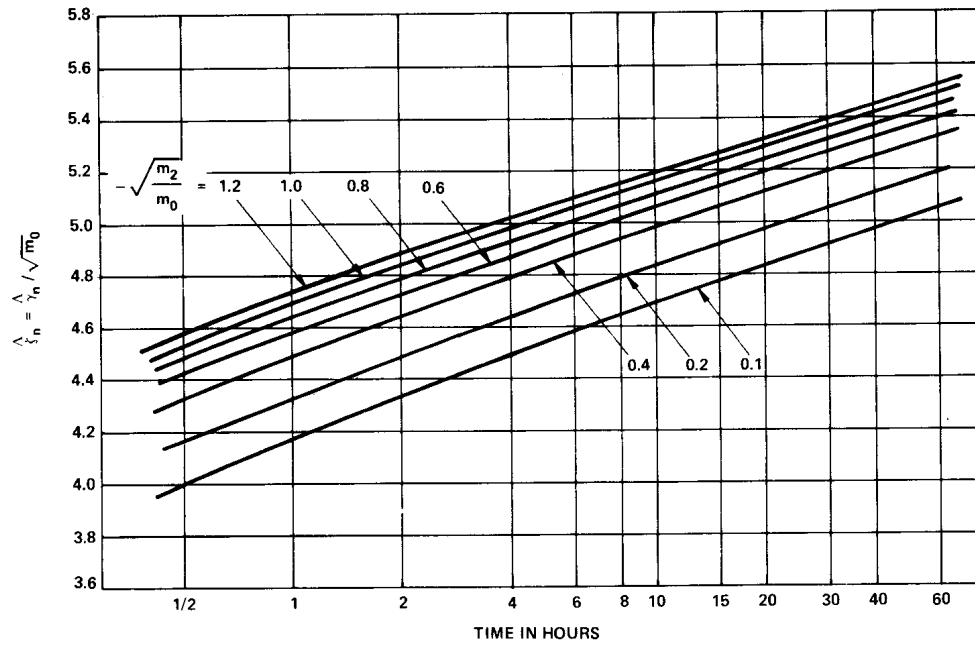


Fig. 5 Extreme value $\hat{\zeta}_n$ (amplitude) as function of time for various values of parameter, $\sqrt{m_2/m_0}$ ($\alpha = 0.01$)

Next, it is shown that the extreme value of the Rayleigh distribution $\hat{\zeta}_n$ given in equation (24) can be derived also by the Cramér's approximation method on extreme statistics [5]. That is, define a random variable $\mathbb{H}$ as

$$\mathbb{H} = n\{1 - F(\zeta_n)\} \quad 0 \leq \mathbb{H} \leq n \quad (31)$$

where $F(\zeta_n)$ is the cumulative distribution function defined in equation (15) where η is ζ_n .

The cumulative distribution function of $\mathbb{H}$ for large n , $H(\theta)$, is then derived as

$$\begin{aligned} \lim_{n \rightarrow \infty} H(\theta) &= \lim_{n \rightarrow \infty} \Pr\left\{F(\zeta_n) \geq 1 - \frac{\theta}{n}\right\} \\ &= \lim_{n \rightarrow \infty} \Pr\left\{\zeta_n \geq F^{-1}\left(1 - \frac{\theta}{n}\right)\right\} \\ &= \lim_{n \rightarrow \infty} \left\{1 - \left(1 - \frac{\theta}{n}\right)^n\right\} = 1 - e^{-\theta} \end{aligned} \quad (32)$$

For the Rayleigh distribution, $\mathbb{H}$ can be written from equations (31) and (15) with $\epsilon = 0$:

$$\mathbb{H} = n \left\{1 - \left(1 - e^{-\frac{\zeta_n^2}{2}}\right)\right\} = ne^{-\frac{\zeta_n^2}{2}} \quad (33)$$

Then, from Equations (32) and (33):

$$\lim_{n \rightarrow \infty} H(\theta) = 1 - \exp\left\{-ne^{-\frac{\zeta_n^2}{2}}\right\} \quad (34)$$

On the other hand, the foregoing cumulative distribution function can be written in terms of a random variable Z_n as follows:

$$\begin{aligned} \lim_{n \rightarrow \infty} H(\theta) &= \Pr\left\{ne^{-\frac{Z_n^2}{2}} \leq ne^{-\frac{\zeta_n^2}{2}}\right\} \\ &= \Pr\{Z_n \geq \zeta_n\} \end{aligned} \quad (35)$$

where Z_n is the random variable defined in Equation (16).

Thus, from equations (34) and (35):

$$\Pr\{Z_n \geq \zeta_n\} = 1 - \exp\left\{-ne^{-\frac{\zeta_n^2}{2}}\right\} \quad \text{for large } n \quad (36)$$

Equation (36) is the asymptotic probability derived based on Cramér's method. Then, by choosing a very small number α for which equation (17) is satisfied, we have

$$1 - \exp\left\{-ne^{-\frac{\zeta_n^2}{2}}\right\} = \alpha \quad (37)$$

Hence, the extreme value $\hat{\zeta}_n$ for small α is obtained by

$$\hat{\zeta}_n = \sqrt{-2 \ln \left\{-\frac{\ln(1 - \alpha)}{n}\right\}} \sim \sqrt{2 \ln \frac{n}{\alpha}} \quad (38)$$

This result agrees with that derived earlier in equation (24), and thus it can be stated that the extreme value derived by Cramér's approximation method agrees with the exact solution derived by applying order statistics if the probability of being exceeded, α , is taken to be small.

Extreme values as a function of time

In the preceding two sections, the extreme values were discussed as a function of number of observations, n , of a random process considered. The number of observations can be obtained directly from the expected number of maxima (per unit time), which is a function of the bandwidth parameter, ϵ . For practical purposes, however, it may be more meaningful to express the extreme value in terms of time rather than as a function of number of observations.

Since the expected number of positive maxima, $N_{m>0}$, is given in equation (9), n can be expressed as a function of time by

$$\begin{aligned} n &= (60)^2 T \bar{N}_{m>0} = (60)^2 \left(\frac{T}{4\pi}\right) \left(\sqrt{\frac{m_2}{m_0}} + \sqrt{\frac{m_4}{m_2}}\right) \\ &= (60)^2 \left(\frac{T}{4\pi}\right) \left(\frac{1 + \sqrt{1 - \epsilon^2}}{\sqrt{1 - \epsilon^2}}\right) \sqrt{\frac{m_2}{m_0}} \end{aligned} \quad (39)$$

where T is time in hours.

Then, from equations (39) and (23), the extreme value can be obtained as a function of time T :

$$\hat{\xi}_n = \sqrt{2 \ln \left\{ \frac{(60)^2 T}{(2\pi)\alpha} \sqrt{\frac{m_2}{m_0}} \right\}} \quad \text{for small } \alpha \quad (40)$$

Figure 5 shows the $\hat{\xi}_n$ -value for $\alpha = 0.01$ as a function of time, T , for various values of $(m_2/m_0)^{1/2}$. As can be seen in the figure, for a given observation time, T , the extreme value $\hat{\xi}_n$ increases with increase with $(m_2/m_0)^{1/2}$.

From equations (30) and (39), the most probable extreme value, $\bar{\xi}_n$, can be expressed also as a function of time T by

$$\bar{\xi}_n = \sqrt{2 \ln \left\{ \frac{(60)^2 T}{2\pi} \sqrt{\frac{m_2}{m_0}} \right\}} \quad (41)$$

It should be noted that the extreme value $\hat{\xi}_n$ as well as $\bar{\xi}_n$ thus expressed in terms of time T is no longer a function of the bandwidth parameter, ϵ , but is a function only of the area and the second moment of the power density spectrum of a random process.

Example of application

As an example of application, the foregoing theory on the prediction of extreme values is applied to ship motion in irregular seas and waves generated in the towing tank.

It should be noted here that the most probable extreme value, $\bar{\xi}_n$, may possibly be used for comparison of the predicted and observed extreme values for the following reason: The observed extreme value is a single sample as compared with other commonly used statistical values such as the average, significant, etc., all of which are evaluated from many outcomes. Suppose only one outcome in n observations is extraordinarily large in comparison with the rest of the sample, this large value is taken as the observed extreme value irrespective of the others. Although, theoretically, the observed extreme value may fall anywhere in the range from 0 to ∞ , it may be reasonable to presume that the extreme value will fall in the vicinity of the value for which the probability density function has a peak, namely $\bar{\xi}_n$.

Figure 6 shows spectra of pitching motion and the 2nd and 4th moments (converted to full-scale ship, $L = 520$ ft) obtained from model experiments carried out on the modified Mariner (Mariner with V-shape bow) in irregular waves representing a Neumann's sea-state 7 (significant wave height 27.3 ft). The spectra are for light-draft condition of the ship and for a 10-knot speed, full scale. The test was conducted for 23.5 min (full scale), and 160 maxima were observed from which the histogram of the pitch motion (double amplitudes) was constructed as shown in Fig. 7. Included also in the figure is the Rayleigh probability density function, the parameter of which was determined by multiplying the area under the spectrum, m_0 , by eight. Good agreement between the histogram and Rayleigh probability density function can be seen in Fig. 7, although the bandwidth parameter of this spectrum, ϵ , is evaluated as 0.378 from the formula given in equation (8).

The number of maxima during the test time of 23.5 min (full scale) can be computed by equation (39), and the result is $n = 172$ as compared with the observed number, 160.

The nondimensional most probable extreme value, $\bar{\xi}_n$, during the test time can be obtained by equation (41). The predicted nondimensional extreme pitch is then 3.20, and hence the extreme pitching motion (double amplitudes), $\bar{\gamma}_n$, becomes $3.20 \sqrt{m_0} \times 2 = 16.3$ deg. On the other hand, the observed extreme value was 18.9 deg (double amplitudes), about 16 percent greater than the predicted value. This result, however, is not surprising, since the probability of occurrence of a value greater than the most probable value is rather high, as was mentioned earlier.

Several other comparisons of extreme values are made for pitching motion of a ship for other test conditions, for ship vertical bow acceleration, and for irregular waves generated in a towing tank. The results are given in Table 1. Included also in

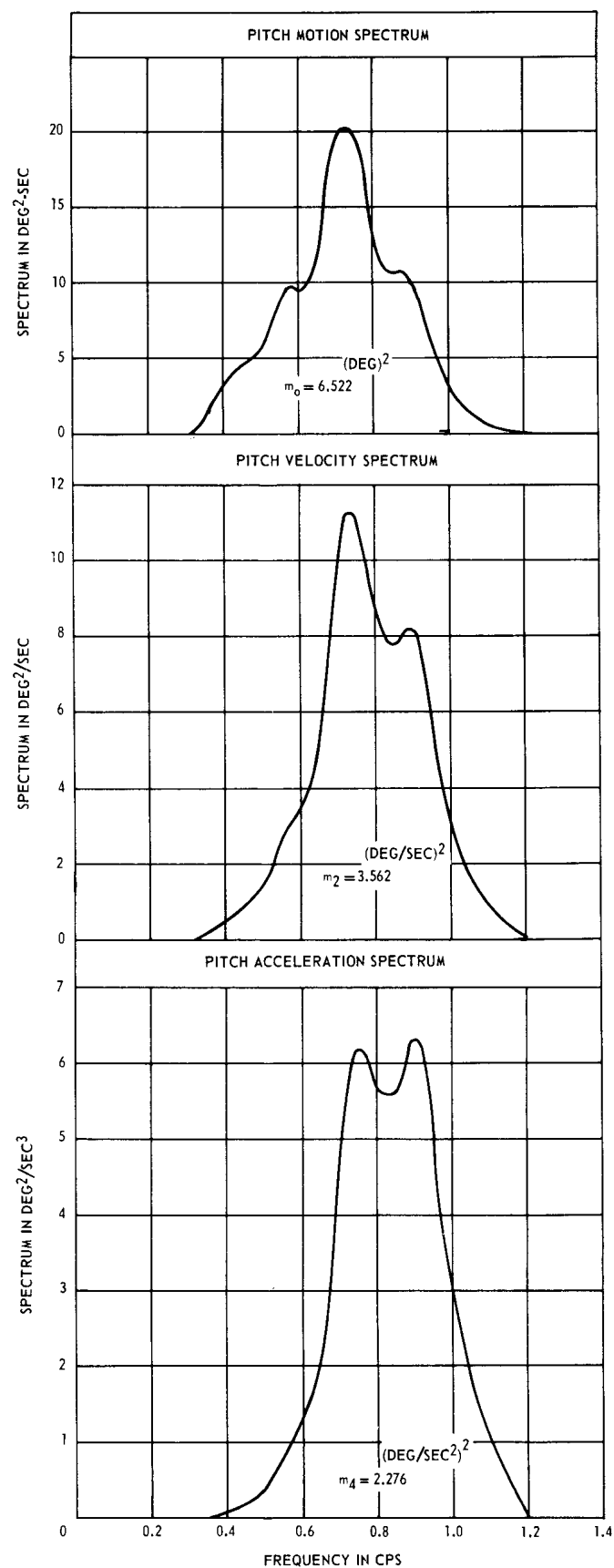


Fig. 6 Pitching motion, velocity, and acceleration, spectra of modified Mariner (see-state 7, significant wave height 27.3 ft, speed 10 knots, light draft, head seas)

the table, for reference, are the dimensional predicted extreme values, $\hat{\gamma}_n(\alpha)$, for $\alpha = 0.10, 0.05$, and 0.01 . As can be seen in the table, the observed extreme values fall fairly well in the vicinity of the most probable values, $\hat{\gamma}_n$, and none of the observed extreme values exceeded the predicted extreme value, $\hat{\gamma}_n$, for α equal to or less than 0.10 .

Table 1 may serve also to provide guidance as to what should be considered appropriate as the extreme value for ship (or marine structure) design consideration. Although the six examples given in the table pertain to the extreme values of ship motions and waves, the theory given in this paper is equally applicable to other phenomena that are directly associated with the safety

of a ship, such as wave-induced bending moment or wave forces acting on a marine structure. For these phenomena, information needed for design is an estimate of the extreme value with a probability, α , of being exceeded rather than the most probable extreme value. Choosing α as 0.01 , for example, implies that only one ship (or marine structure) in 100 sister ships (structures) operating under statistically identical environments may suffer from a bending moment, force, etc., greater than the predicted value in a given period of time. In other words, α may be interpreted as providing a probability of assurance. If greater assurance for safety is required, one may choose a smaller α depending on the particular situation.

The agreement between the predicted most probable and observed extreme values is satisfactory in the examples given in Table 1. However, the chance that the extreme value will exceed the most probable value is rather high; the probability is 0.63 , theoretically, for a random process having a narrow-band spectrum. Hence, the most probable extreme value may not be appropriate for consideration as the design value. On the other hand, the extreme value $\hat{\gamma}_n$ for $\alpha = 0.01$ that is approximately 35 to 40 percent (in these examples) greater than the most probable extreme value appears to be appropriate from a comparison with the observed extreme value.

Conclusions

Formulas to estimate the extreme value of a stationary random process having an arbitrary bandwidth spectrum are derived by applying order statistics both for a given number of observations and for a given time during which the sea environment is constant. The most probable extreme value and the extreme value for a preassigned probability of being exceeded are obtained. Conclusions derived from this study are as follows:

- The extreme value that will occur in n cycles, $\hat{\xi}_n$ (nondimensional), can be obtained from equation (20) for a preassigned probability of being exceeded, α . In particular, for a bandwidth parameter, ϵ , less than 0.9 , the extreme value is obtained by equation (23). The dimensional extreme value (amplitude) is obtained by multiplying $\hat{\xi}_n$ by $\sqrt{m_0}$, where m_0 is the area under the spectral density function of the random process.
- The most probable extreme value, $\hat{\xi}_n$, the value for which the probability density function becomes maximum, is obtained by equation (30).
- For the same number of observations, n , the effect of the bandwidth parameter, ϵ , on the extreme value is small for ϵ less than 0.90 .
- For a random process having a narrow-band spectrum (ϵ

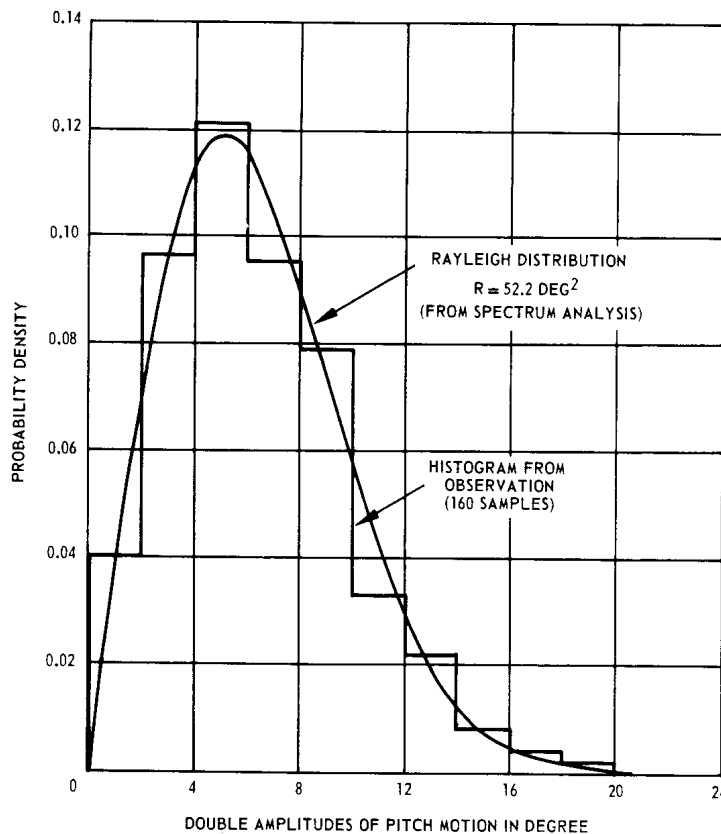


Fig. 7 Observed histogram and theoretical Rayleigh distribution of pitching motion (double amplitudes)

Table 1 Comparison of predicted and observed extreme values

	TEST TIME, min	BAND- WIDTH PARAMETER ϵ	NUMBER OF MAXIMA		OBSERVED EXTREME VALUE, double amplitudes	Most probable extreme value, $\hat{\gamma}_n$	PREDICTED EXTREME VALUE USING OBSERVED SPECTRUM		
			ob- served	esti- mated			Extreme value, $\hat{\gamma}_n$		
							$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
Pitching motion (Sea-state 7, 10-knot speed)	23.5	0.378	160	172	18.9 (deg)	16.3	19.7	20.6	22.5
Pitching motion (Sea-state 6, 15-knot speed)	24.8	0.350	233	211	11.2 (deg)	12.1	14.4	15.1	16.5
Bow acceleration (Sea-state 7, 10-knot speed)	37.4	0.231	360	326	1.78 (g)	1.75	2.07	2.16	2.35
Generated waves (Sea-state 5)	27.5	0.513	177	186	22.3 (ft)	20.6	24.8	26.0	28.4
Generated waves (Sea-state 6)	24.8	0.616	282	308	32.0 (ft)	33.3	39.6	41.3	45.0
Generated waves (Sea-State 7)	32.9	0.598	213	220	48.7 (ft)	45.1	54.2	56.6	61.9

= 0), the peak values have a Rayleigh probability distribution, and its extreme values are given by equations (24) and (25). The extreme value $\hat{\xi}_n$ thus derived is much greater than the most probable extreme value, $\tilde{\xi}_n$.

- For a random process having a narrow-band spectrum, the extreme value derived by applying Cramér's approximation method agrees with the exact solution derived by applying order statistics if the probability of being exceeded, α , is chosen to be small.

- The extreme value for a given period of time is evaluated by equations (40) and (41). The extreme value expressed in terms of time is no longer a function of the bandwidth parameter, ϵ , but is a function only of the area and the 2nd moment of the spectral density function of a random process.

- Comparisons between the predicted and observed values were made for the pitching motion and bow acceleration of a ship in irregular seas and waves generated in a towing tank. It was found that the extreme values observed in model experiments fell fairly well in the vicinity of the predicted most probable values.

- As far as experimental results are concerned, none of the observed extreme values exceeded the predicted extreme value for α equal to or less than 0.10. However, the extreme value for $\alpha = 0.01$, which is approximately 25 to 40 percent (depending on the given period of time) greater than the most probable extreme value, appears to be appropriate for ship (or marine structure) design consideration.

Acknowledgment

The author wishes to express his appreciation to Dr. W. E. Cummins for his valuable comments offered during the review of this paper. This work was carried out under the General Hydromechanics Research Program, U. S. Navy.

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Appendix

Expected number of maxima with positive value

The expected number of maxima with positive value per unit time, denoted as $\tilde{N}_{m>0}$, is given by equation (5) in the text.

That is

$$\tilde{N}_{m>0} = \int_0^\infty \int_{-\infty}^0 |\dot{x}(t)| f(x, 0, \dot{x}) d\dot{x} dx \quad (42)$$

where $f(x, 0, \dot{x})$ is the joint probability density function of the random processes $\{x(t), 0, \dot{x}(t)\}$, the covariance matrix of which is given in equation (2). By using the notation given in equation (2), we have

$$|\Sigma| = m_2(m_0m_4 - m_2^2) = m_2\Delta \quad (43)$$

and

$$\mathbf{x}'\Sigma^{-1}\mathbf{x} = \frac{m_4x^2 + 2m_2x\dot{x} + m_0\dot{x}^2}{\Delta} \quad (44)$$

where

$$\Delta = m_0m_4 - m_2^2$$

Then, the probability density function $f(x, 0, \dot{x})$ is written by

$$f(x, 0, \dot{x}) = \frac{1}{(2\pi)^{3/2}(\Delta)^{1/2}} \exp\left\{-\frac{m_4x^2 + 2m_2x\dot{x} + m_0\dot{x}^2}{2\Delta}\right\} \quad (45)$$

From equations (42) and (45), the integration with respect to x yields

$$\tilde{N}_{m>0} = \frac{1}{(2\pi)\sqrt{m_2m_4}} \int_{-\infty}^0 |\dot{x}| e^{-\frac{\dot{x}^2}{2m_4}} \left\{1 - \Phi\left(\frac{m_2}{\sqrt{m_4\Delta}} \dot{x}\right)\right\} d\dot{x} \quad (46)$$

By taking into account that $\dot{X} < 0$ for the present problem, the integration of equation (46) becomes

$$\begin{aligned} & \int_{-\infty}^0 |\dot{x}| e^{-\frac{\dot{x}^2}{2m_4}} \left\{1 - \Phi\left(\frac{m_2}{\sqrt{m_4\Delta}} \dot{x}\right)\right\} d\dot{x} \\ &= \int_0^\infty \dot{x} e^{-\frac{\dot{x}^2}{2m_4}} d\dot{x} + \int_{-\infty}^0 \dot{x} e^{-\frac{\dot{x}^2}{2m_4}} \Phi\left(\frac{m_2}{\sqrt{m_4\Delta}} \dot{x}\right) d\dot{x} \\ &= \frac{1}{2} \left(m_4 + m_2 \sqrt{\frac{m_4}{m_0}}\right) \end{aligned} \quad (47)$$

Hence, $\tilde{N}_{m>0}$ becomes

$$\begin{aligned} \tilde{N}_{m>0} &= \frac{1}{(4\pi)\sqrt{m_2m_4}} \left(m_4 + m_2 \sqrt{\frac{m_4}{m_0}}\right) = \left(\frac{1}{4\pi}\right) \\ &\quad \times \left(\sqrt{\frac{m_2}{m_0}} + \sqrt{\frac{m_4}{m_2}}\right) \end{aligned} \quad (48)$$